Vinita Deshmukh (Ph.D)

(+33) 0754501550 | vinita.deshmukh@univie.ac.at | https://orcid.org/0009-0009-6012-3348 | https://github.com/Vinita-D | vinita-deshmukh-b300b9197

Profile

Weather and climate researcher with almost 5 years of experience diagnosing large-scale circulation patterns that drive European weather and climate extremes. Working understanding of weather and climate variability, climate and weather extremes. Strong programming and analytical skills (Python, HPC/Linux, FORTRAN) with extensive experience in weather and climate modeling and handling large climate datasets and model outputs. Experienced in international research collaborations and interdisciplinary projects, building and leading community exchange initiatives (public workshops, a recurring discussion group) alongside a academic research and maintaining publication record.

Research Experience

Institut für Meteorologie und Geophysik, Universität Wien, PostDoc

Vienna, Austria

Quantifying Heat and moisture transport in weather extremes

02/2025 - 09/2026

- Developed a Lagrangian moist static energy framework to attribute heatwave intensity to surface-flux and dynamical contributions.
- Quantifying contribution from fluxes and advected airmass
- Validated the diagnostic framework against Storm Boris
- Quantifying contribution from fluxes and advected airmass
- Contributed to the Edu4ClimAte Horizon Europe project; manuscripts submitted to WCD.

Laboratoire de Météorologie Dynamique (LMD), CNRS. - Ph.D. thesis

Paris, France

Dynamics of Atmospheric blocking and its link with warmer temperature extremes.

10/2021 - 12/2024

- Conducted long-term simulation of a dry General Circulation Model (GCM) forced to ERA5 climatology.
- Evaluated long-term GCM simulations against ERA5 reanalysis to identify, characterise and attribute systematic model biases in the representation of atmospheric blocking.
- Employing Eulerian and Lagrangian framework to understand blocking in the dry model.
- Supervisors: Dr. Gwendal. Rivière (LMD-ENS), Dr. Sébastien Fromang (LSCE).

EDIPI Ph.D Exchange - Imperial College

London (UK)

Influence of latent heat fluxes from the Gulf Stream on Euro-Atlantic blocking circulation.

02/2023-05/2023

- Collaborative projects within EU-Research Framework - H2020 MSCA, EDIPI
- A sensitivity experiment using ECMWF-IFS data with suppressed latent heat fluxes is utilised.
- Analyzed two blocking cases in detail: the 2010 British Isles cold spell and the 2019 European heatwave.
- In collaboration with Dr Arnaud Cazja, Jamie Mathews and Dr. Gwendal Rivière.

EDIPI Ph.D Exchange - Uppsala University

Uppsala (Sweden)

Investigating Link Between Blocking Characteristics and Heatwaves.

05/2024 - 08/2024

- Developed Quasi-Lagrangian technique of detecting heatwaves.
- Establishing statistical link between blocking and heatwave characteristic.
- Quantified the statistical link between blocking characteristics and mid-latitude heatwaves, including persistence and recurrence effects on heatwave onset.
- Applied multivariate regression methods to link large-scale atmospheric patterns to temperature extremes.
- In collaboration with Dr. Gabriele Messori, Uppsala University, Sweden.

Indian Institute of Tropical Meteorology - Master's thesis

Pune, India

Understanding Low-Pressure System in monsoon trough region from the Bay of Bengal.

01/2020 - 06/2020

- Understanding Convective characteristic associated with different phases of LPSs, Depression, and Deep Depression.
- Dataset from Remote sensing instruments like space-borne radars on board with the TRMM satellite.
- Supervised by Dr. Subrata Das Kumar, IITM and Abhishek Jha, IITM

Education

Ph.D in Atmospheric dynamics: École normale supérieure - Université PSL

France

Ph.D. Student at Laboratoire de Météorologie Dynamique (LMD).

10/2021 - 10/2024

- Under the H2020 MSCA-ITN, https://edipi-itn.eu/

M.Sc: Dept. of Atmospheric Science, Savitribai Phule Pune University (SPPU)

India

Master's in Atmospheric Science in collaboration with IITM

06/2019 - 06/2021

Undergraduate: Ferguson College, Pune

India

Major in Physics

06/2015 - 06/2018

Technical Skills

Atmospheric dynamics, large-scale weather and climate analysis.

- Programming: Python (xarray, numpy, pandas, scipy, scikit-learn, matplotlib, Cartopy), Fortran; Git
- Numerical modelling using HPC/Linux: WRF, GCM, ICON (training); Lagrangian modelling (FLEXPART, LAGRANTO, LagranTraj)
- Data: ERA5 reanalysis, ECMWF-IFS output, satellite observations (TRMM); NetCDF, GRIB

Training events/Schools

EDIPI Marie-Curie-ITN training events

Europe

Training, experiments and social outreach, as a part of the EDIPI project.

- Statistical concepts in extreme weather events and modelling, Research Ethics, Uppsala University Sweden.
- Climate change attribution, tipping points, statistical Methods, research communication strategies, Paris, France.
- Idealised modelling, Socio-Ecological Systems, and Numerical Weather Forecasting, Brussels, Belgium.
- Climate model simulation (ICON model), Climate Risk Assessment and Communication, Presentations by Industry Experts, Climate Education session, Karlsruhe, Germany

Publications

* V. Deshmukh;, Gwendal Rivière, Sébastien Fromang, and Marion Saint-Lu. How does a dry general circulation model represent atmospheric blocking? *Published in AMS- JAS* `https://doi.org/10.1175/JAS-D-24-0048.1`, 11 2024

* V Deshmukh, Andreas Stohl, and Marina Dütsch. Surface flux contributions to mediterranean heatwaves: a new lagrangian diagnostic. *Submitted to Weather and Climate Dynamics (WCD)*, 2026

* V. Deshmukh;, G. Messori, and G. Rivière. The link between blocking characteristics and heatwave. *[Revision - submitted to ERL journal]*

* V. Deshmukh. Dynamics of atmospheric blocking and its link to extreme temperature events. *PhD dissertation* `:https://doi.org/10.70675/76c54928zf116z4c5fz86d3z01e937560f4e`, 12 2024

* S K Das; A K Jha; U V MuraliKrishna; V Deshmukh. Radar observed convective storm characteristics at the eastern edge of the indian summer monsoon trough. `https://essopenarchive.org/doi/pdf/10.22541/essoar.176030779.96371703/v1`, 61:3633–3652, 10 2023

Talks & Posters

2026 *Presentation*, EGU General Assembly, Vienna, Austria (*talk*).
2026 *WP2 presentation*, Edu4ClimAte project meeting (*talk*).
2025 *Presentation*, Stormtrack workshop, Rosendal, Norway (*poster*).
2024 *US CLIVAR Workshop on Blocking and Extreme Weather in a Changing Climate*, Boulder, CO, USA (*poster*).
2024 *Visiting seminar*, University of Chicago, USA (*talk*).
2023 *Presentation*, EGU General Assembly, Vienna, Austria (*talk*).
2023 *IUGG General Assembly*, Berlin, Germany (*talk*).
2023 *Visiting seminar*, Imperial College London, UK (*talk*).
2023 *Visiting seminar*, KIT, Karlsruhe, Germany (*talk*).
2023 *Luval seminar*, Uppsala University, Sweden (*talk*).
2022 *FDSE summer school*, Paris, France (*poster*).

Responsibilities and Outreach

2026 *Organised workshop, "How to read storms", Vienna Klima Biennale*Vienna, Austria
2025 *Teaching assistant, Atmospheric Thermodynamics*Dept. of Meteorology and Geophysics, Vienna, Austria
2025 *Co-organiser, Edu4ClimAte Autumn School on atmospheric transport modelling*Crete, Greece
2025 *Founder and organiser, weekly weather discussion group*Universität Wien, Vienna, Austria
2023 *Elected Early-Career Researcher Representative*IUGG General Assembly, Berlin, Germany
2022 *Initiated and led "Café des Géosciences" programme*LMD-ENS, Paris, France

Languages

Marathi (native), English (fluent), Hindi (fluent), French (A2) , German (A1)